

Where in the US would I want to be hospitalized if I wanted the lowest risk of C. difficile infection?

Analysis of C. difficile Standardized Infection Ratios across US hospitals

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Introduction

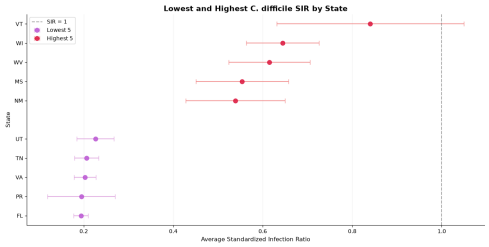
Hospital-acquired infections are infections that develop during or as a result of receiving healthcare treatments. The topic has been increasing in relevance within the sector, as they can worsen patient outcomes through a multitude of complications in both patient care and hospital management [3]. One major example is Clostridioides difficile, a bacterium that affects the gastrointestinal tract. Common clinical manifestations are diarrhea and colitis [5]. C. difficile is one of the six HAIs in the dataset used in this project, all measured through SIR [2]. C. difficile was selected for further analysis mainly due to the presence of valid observations within the dataset, both numerically and geographically, making it suitable for state-level comparison. To compare infection outcomes, the analysis uses the Standardized Infection Ratio (SIR), which compares observed infections with predicted infections [2]. An SIR number below 1 indicates fewer infections than predicted, while one above 1 indicates more infections than predicted [2]. This project therefore investigates C. difficile outcomes across states, with the aim of identifying which state has the lowest average C. difficile SIR

Data sample and method

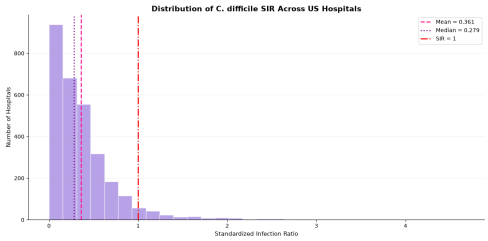
The analysis was based on the CMS Healthcare Associated Infections Hospital dataset, which contained 4,790 hospitals, 56 states and territories, and 6 infection types with SIR data [2]. Before the analysis, the Score variable was cleaned by turning non-numerical entries into missing values. For the main analysis, the dataset was filtered to include only SIR measurements specifically for C. difficile, and any missing scores were removed. To compare geographic differences, hospital C. difficile SIR values were grouped by state, and the mean SIR was calculated. The mean represented the average hospital SIR within each state. If states had fewer than 10 valid measurements, they were excluded in order to avoid comparisons based on samples that were too small. This maintained 49 states and territories for the comparison. Next, variation between hospitals within each state was assessed using the sample standard deviation. This helped to measure how spread out hospital values were around the state mean. Afterwards, the uncertainty of each state mean was estimated using the standard error, calculated as the SD divided by the squaree root of N, where a higher N indicated lower uncertainty [4]. At this point, the lowest and highest state mean SIR values were identified, and the difference between them was calculated. The results were then compared for further interpretation. Finally, Pearson correlation coefficients were calculated to examine how the state average C. difficile SIR compared with the other infection types [1]. The strongest absolute correlation was selected for the final scatter plot.

Results and discussion

The analysis showed that Florida had the lowest average C. difficile SIR, with a mean of 0.194 ± 0.016, while Vermont had the highest average, with a mean of 0.840 ± 0.209. This highlighted the substantial difference of 0.646. The difference is relevant, as the uncertainty ranges of Florida and Vermont did not overlap, making the observed difference larger than the estimated uncertainty. The state ranking plot further showed clear variation between states (Figure 1, left). The lowest and highest states were visibly separated. Error bars were added to represent the standard error.



(a) Lowest and highest average C. difficile SIR values by state. Error bars represent the standard error of the state mean.



(b) Distribution of C. difficile SIR values across hospitals. Vertical lines indicate the mean, median, and SIR = 1.

Figure 1: C. difficile SIR analysis across US hospitals and states.

At the hospital level, the distribution of C. difficile SIR values was right-skewed (Figure 1, right), showing a mean of 0.361 and a median of 0.279. The mean being higher than the median suggests that some high SIR values pulled the average upward. The additional calculation was needed, as the mean alone does not fully describe all hospitals. The correlation analysis was used to examine whether states with relatively low C. difficile SIR also tended to have a low SIR for another infection [1]. It was calculated between the state average C. difficile SIR and the other infection types. The strongest relationship was observed between C. difficile and SSI following colon surgery and was selected. The coefficient r equaled 0.278, indicating a weak positive relationship [1] (Figure 2).

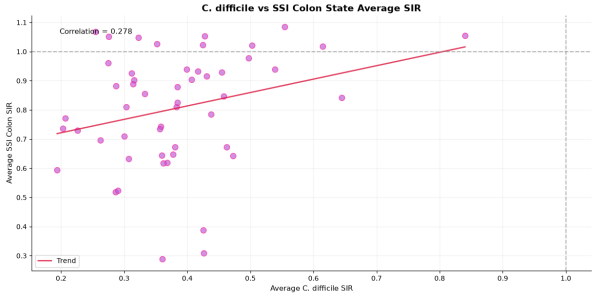


Figure 2: Relationship between state average C. difficile and SSI Colon SIR. Each point represents a state and the line shows the overall trend.

This meant that states with a higher average C. difficile SIR tended to also have a slightly higher SSI colon SIR value. However, because the correlation was weak, it did not imply that one infection causes the other. Overall, the analysis indicates that Florida had the lowest average C. difficile SIR, suggesting better outcomes at the state level. However, the results represent aggregated hospital SIR values rather than individual patient risk.

References

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